Implementation and Comparative Analysis of Social Media Sentiment Classification: LSTM RNN vs. Fine-Tuned Transformer Models (BERT & RoBERTa)

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ARTICLE INFO

Keywords:
NLP Classification
Applied Deep Learning
RNN LSTM
Transformers
BERT
RoBERTa

ABSTRACT

This paper addresses the increasing prevalence of cyberbullying in social media and proposes a method to detect harmful content through sentiment analysis and classification. With a dataset of over 47,000 social media threads, the study explores various methods, including RNN (LSTM), BERT, and RoBERTa, to classify cyberbullying. The RNN (LSTM) model with Word2Vec achieved an accuracy of 81.93%, showing effective performance in identifying cyberbullying but indicating room for improvement in distinguishing between different types of cyberbullying. BERT and RoBERTa models, which leveraged pre-trained transformers, outperformed the RNN (LSTM) model, with BERT achieving an accuracy of 85.78% and RoBERTa achieving 85.25%. Both transformer models demonstrated higher classification accuracy, though RoBERTa exhibited a more balanced precision and recall across categories.

Although all models showed promising capabilities for cyberbullying detection, transformer models, especially BERT and RoBERTa, offered superior performance in accurately classifying social media posts. BERT showed the highest accuracy, with minor Type I errors in misclassifying non-bullying threads as bullying. RoBERTa, while computationally more expensive, offered slightly better balanced performance across various metrics. These findings suggest that both RNN LSTM and pre-trained transformer models, BERT and RoBERTa, offer a robust framework for detecting cyberbullying in real-world social media data. For future research, further refinement in pre-processing of imbalanced categories and implementation of Adapters and LoRA could potentially improve performance.

1. Introduction

Cyberbullying has become more prevalent with the increasing use of social media across all age groups. A vast majority of people use social media for daily communication, and this ubiquity means that cyberbullying in this form can affect anyone. Given the ability to hide behind a screen, these personal attacks are much more undetectable, inconsequential, frequent, and unavoidable than traditional bullying. While it is difficult to protect people from the internet, there is a wealth of data available, especially in social media speech and tweets. By understanding the different types of discrimination—gender, religion, age, ethnicity, and others—we can teach our model to categorize and flag these threads to build a system that can combat the consequences of discriminatory speech.

As displayed in the class distribution in Figure 1, this paper uses a dataset containing more than 47,000 social media threads (∼7,800 per classification), each labeled with its category for cyberbullying. The dataset has a balanced representation, with duplicates removed and no null values.

By understanding the sentiment of social media posts and classifying them accordingly, we can create a data-driven method to better understand how we can combat these issues. Given the text-driven nature of cyberbullying, we can leverage this data to design models that detect harmful content more effectively.

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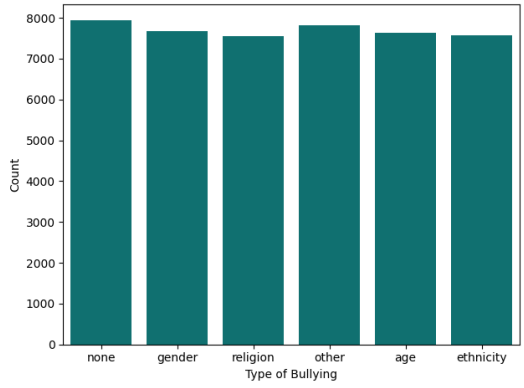

Figure 1: Sentiment Distribution of Dataset

This paper will follow the structure outlined below:

- **Literature Review:** Examine existing techniques and tools for sentiment analysis, laying the groundwork for this study.
- **Methodology:** Propose a novel deep learning approach leveraging pre-trained transformer models to classify cyberbullying.
- **Results and Discussion:** Present the findings from the model's performance, analyzing its effectiveness in optimizing classification.

- **Conclusion:** Reflect on the real-world implications of the proposed method and explore opportunities for further research and model improvement.

2. Literature Review

This section of the report is structured to better understand current methods used for NLP sentiment analysis, mainly in RNN-LSTM and fine-tuning transformer-based models, BERT and RoBERTa, and how they are evaluated.

2.1. Sentiment Analysis using NLP

Sentiment analysis is an emerging field in text mining. It is an NLP application that identifies a text corpus's emotional or sentimental tone or opinion—usually emotions or attitudes toward a topic in varying classes, essentially identifying and categorizing opinions expressed in a piece of text over different social media platforms [1].

Recurrent Neural Networks (RNN) are a form of ANN that can memorize length sequences of input patterns by capturing connections between sequential data types [2]. To avoid exploding and vanishing gradients, Long Short Term Memory (LSTM) networks are a type of RNN designed to mitigate these issues through gating mechanisms [3]. While RNNs are limited to looking back in time for approximately ten timesteps [4], LSTM networks are capable of learning more than 1,000 timesteps, depending on the complexity of the network [5]. This makes LSTM-RNNs a very powerful dynamic classifier for sequential data [6]. Its implementation with NLP has been seen in studies surrounding natural language inference [7] and sentiment analysis tasks by integrating Word2Vec for hotel reviews [8].

Word2Vec is a model for representing words in vector space, measuring the quality of these representations in word similarity tasks to assess syntactic and semantic word similarities. There are two models that allow transforming a word into a numerical vector: Continuous Bag-of-Words (CBOW) and Skip-Gram. CBOW predicts a center word using the words around it, while Skip-Gram predicts words around a center word [9]. Using a pre-trained Word2Vec embedding method, it is possible to integrate Word2Vec into an RNN-LSTM task [10], as used in this paper.

Bidirectional Encoder Representations from Transformers (BERT) is designed to pre-train deep bidirectional representations from unlabeled text by jointly conditioning on both left and right context in all layers. This enables BERT to be pre-trained and fine-tuned with just one additional output layer to create state-of-the-art models for a wide range of tasks [11]. Some examples in recent research include classifying mobile app star ratings based on user reviews [**?**] and sentiment analysis for Vietnamese reviews, outperforming other models that use GloVe and FastText for word embeddings.

The Robustly Optimized BERT Approach (RoBERTa) is a BERT variant introduced by Facebook AI Research as an improvement over BERT, removing the Next Sentence Prediction objective and using only the Masked Language Modeling objective. RoBERTa uses larger mini-batches and learning rates to learn more effectively with increased training data, while also employing dynamic masking to introduce variability and help models learn more flexible patterns [12].

Other methods explored outside of this paper to improve the performance of pretrained models are Adapter and Low-Rank Adaptation (LoRA). An adapter is a small module inserted into pre-trained models like BERT to enable task-specific fine-tuning without significantly modifying the original model's weights, commonly used in transfer learning [13]. LoRA tackles the challenge of full fine-tuning with larger models by freezing the pre-trained model weights and injecting trainable rank decomposition matrices into each layer of the Transformer architecture, reducing the number of trainable parameters for downstream tasks [14].

2.2. Evaluation Metrics for Classification Models

Evaluation of classification models differs based on the input dataset and the task at hand, whether it is balanced or imbalanced. Upon research of 16 different metrics, [15] found that the selected dataset should be pre-processed to be reliably classified by the classification model. To understand how the classification is correctly performed, confusion matrices are effective summary tables that compare the expected and actual class labels, capturing the true positives, true negatives, false positives, and false negatives [16]. The trade-off between the true positive rate and false positive rate for a predictive model utilizing various probability thresholds is summarized by Receiver Operating Characteristic (ROC) Curves [17].

The choice of evaluation metrics and reasoning behind are further discussed in Section 3.3.

3. Methodology

This section describes the methodology for evaluating the performance of three models—BERT, RoBERTa, and RNN (LSTM)—on a sentiment classification task. Each model will be fine-tuned and evaluated using a structured process. LR, Epochs are optimized based on suggestions in prior research [11], followed by hyperparameter tuning and performance comparison.

3.1. Data Pre-Processing and Model Preparation

For the most part, to ensure consistency, the same preprocessing method is used for BERT and RoBERTa. Whereas, a slightly different method is used for RNN (LSTM).

Figure 2 shows that the data is limited to threads containing up to 30 words, as the distribution predominantly falls within this range, with a few outliers exceeding 100 words. Transformer models handle these variations by applying padding to shorter sequences and truncation to longer sequences, ensuring inputs fit the specified maximum sequence length.

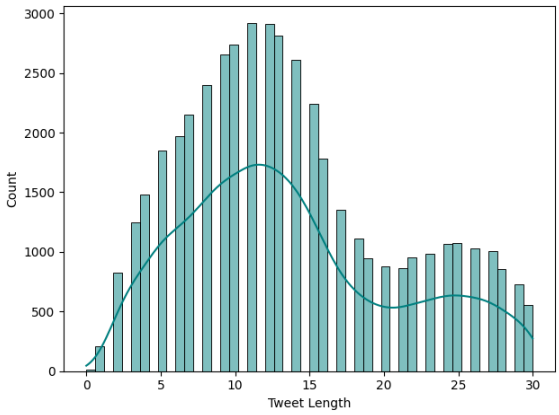

Figure 2: Text Length Distribution for Threads

3.1.1. RNN (LSTM)

A bidirectional LSTM with an attention mechanism is designed for sentiment classification. The data is preprocessed through various steps, including tokenization and embedding using a pre-trained Word2Vec model. This ensures a consistent approach for validation and testing while preserving compatibility across models.

To handle the sequence modeling and classification, two custom classes are implemented:

- **Attention:** This class implements the attention mechanism, which assigns weights to different parts of the input sequence to focus on the most relevant features for generating the output.
- **LSTMClassifier:** This class integrates the attention mechanism and defines the overall model structure. It consists of an embedding layer, a bidirectional LSTM for sequence processing, and a fully connected layer for prediction. The attention mechanism processes the hidden states of the LSTM to derive a context vector, which is then used for classification.

The model is trained and validated using a stratified split of the data to maintain balanced class distributions. To enhance training performance, oversampling is applied to address class imbalances in the training set. The classification performance is evaluated on the test data, ensuring robust comparison across models.

3.1.2. BERT and RoBERTa

For BERT and RoBERTa, text preprocessing typically involves several steps to clean and standardize the input data. Initially, emojis are removed using the `remove_emojis` function, which converts emojis to text representations and removes the emoji characters. Next, URLs are eliminated using a regular expression pattern (`remove_urls`), which removes any web links from the text. HTML tags are also removed with a regular expression (`remove_html`) to ensure that the model only receives relevant content. Punctuation is stripped using the `remove_punctuations` function, which ensures that the text is free from non-alphabetical characters that could interfere with tokenization. Additionally, stopwords are filtered out using the `remove_stopwords` function, which removes commonly occurring but less informative words like "the" and "and". To standardize the words, stemming (`stem_words`) and lemmatization (`lemmatize_words`) are applied, reducing words to their root forms. After these preprocessing steps, the text is ready for tokenization using BERT and RoBERTa's respective tokenizers, which convert the text into subword units suitable for further processing by the models. This ensures that the models receive clean, consistent input data for fine-tuning.

The objective of the tokenizers for BERT and RoBERTa is to encode the text and return tensors as 'pt', assigning `token_ids` and `attention_masks` relevant to both. The training samples and validation samples are then created with a 2:8 ratio, providing around 36,960 training samples and 9,241 validation samples. Now that the data is prepared, the next part discusses the LR, Epochs used to find the optimal solution.

The remaining pre-processing and tokenization for each model is computed in the following sections.

3.2. Hyperparameter Tuning

To ensure optimal performance, hyperparameter tuning is performed across all three models. The tuning strategies are tailored to the specific architecture of each model:

- **RNN (LSTM)**: Key hyperparameters, including the number of hidden units, learning rate, dropout rate, and the number of epochs, are tuned to achieve the best results.
- **BERT and RoBERTa**: Grid search is applied to explore variations in learning rates and the number of epochs. Baseline configurations are guided by recommendations from the literature and further adjusted based on validation performance.

The validation set is used to evaluate the performance of each configuration, and the best hyperparameters are selected for final testing.

3.2.1. RNN (LSTM)

RNN with LSTM units and an attention mechanism is implemented for sentiment classification. This model is particularly effective for sequential data, as it can capture long-term dependencies in text. The architecture includes:

The LSTM model is configured with 128 hidden units to balance model complexity and performance. A dropout rate of 0.5 is applied to mitigate overfitting by deactivating neurons randomly during training. The learning rate is set to 0.001 (and tested with other values 0.003, 0.0001), optimized through experiments. The number of epochs ranges until 10, determined by employing early stopping to prevent overfitting, as seen in Figure 3. This configuration ensures the model effectively captures sequential dependencies in text while maintaining robust generalization across datasets.

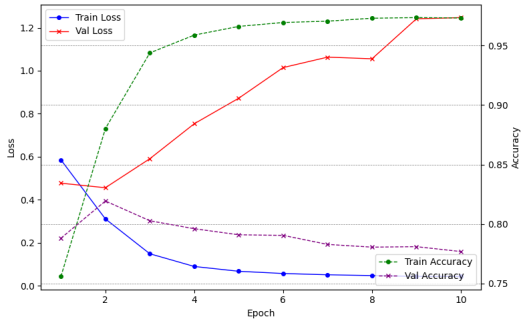

Figure 3: RNN Early Stopping for Best Model: Training and Validation Loss and Accuracy

The LSTM classifier employs pre-trained Word2Vec embeddings for feature representation. A custom attention mechanism is used to assign varying levels of importance to different parts of the input sequence, improving interpretability and accuracy. This is particularly useful for nuanced sentiment analysis, where the context plays a critical role. For example, in the sentence, "I hate that I love you," the attention mechanism can help the model focus on contrasting sentiment cues, enabling more accurate classification.

3.2.2. BERT

BERT (Bidirectional Encoder Representations from Transformers) is fine-tuned with all gradients enabled, ensuring that weights in every layer are updated. The experiments are conducted with learning rates of 2×10^{-5} and 3×10^{-5}, a batch size of 16, and a maximum sequence length of 128. Fine-tuning is performed over 2 or 3 epochs, involving a total of 109,491,468 parameters. In total, 4 trials are conducted to identify the best-performing configuration, primarily based on accuracy, while also evaluating the performance of each one-vs-rest (OVR) classification.

3.2.3. RoBERTa

RoBERTa (Robustly Optimized BERT Approach) follows a similar fine-tuning process, adapted to its enhanced pretraining strategy. The experiments use learning rates of 2×10^{-5}, 3×10^{-5}, and 5×10^{-5}, with a batch size of 16. The maximum sequence length is set to 128, with additional tests conducted at lengths of 64 and 32. Fine-tuning is performed over 2 or 3 epochs, involving a total of 125,245,452 parameters. Six trials are conducted, and the best-performing model is selected for comparison.

3.3. Evaluation Metrics

The models are evaluated using the following metrics:

- **Accuracy:** Measures the overall correctness of predictions.
- **Precision, Recall, and AUROC:** Evaluated per class and averaged using macro and weighted averages, displayed in the classification report for each model.
- **Confusion Matrix:** Visualizes the distribution of correct and incorrect predictions for each class.

The methodology involves fine-tuning BERT, RoBERTa, and RNN (LSTM) on the same sentiment classification dataset. LR, Epochs are optimized using a manual grid search from literature references. The final models are evaluated on the test set using accuracy, precision, recall, AUROC, and confusion matrix, understanding how these metrics matter for the specific context of cyberbullying.

4. Results and Discussions

This section will cover the model performance based on the evaluation metrics mentioned above. The initial model tested is RNN (LSTM) as a base, followed by more complex models like BERT and RoBERTa, which evaluate the accuracy, classification report, and confusion matrix for each.

4.1. RNN Model Performance

The RNN model was evaluated with various learning rates, with the best performance observed at a learning rate of $lr = 0.001$. The model was trained for 10 epochs, but an early stopping mechanism was implemented to mitigate the risk of overfitting. This mechanism selects the best model based on the lowest validation loss achieved at a specific epoch, ensuring that the model generalizes well to unseen data rather than simply memorizing the training set.

Table 1
Classification Report for RNN LSTM

	precision	recall	f1-score	support
0	0.58	0.55	0.57	1587
1	0.83	0.89	0.86	1548
2	0.93	0.95	0.94	1519
3	0.96	0.97	0.96	1528
4	0.96	0.97	0.96	1541
5	0.64	0.61	0.62	1563
accuracy			0.82	9286
macro avg	0.82	0.82	0.82	9286
weighted avg	0.82	0.82	0.82	9286

The LSTM model achieved an accuracy of 81.93%, which indicates that the model effectively classifies most cyberbullying-related tweets. The high accuracy suggests that the model is able to identify the relevant features associated with cyberbullying, even in a noisy, real-world dataset. However, there is room for improvement, particularly in distinguishing between closely related categories. The model's performance could be enhanced by experimenting with additional hyperparameters, such as the number of LSTM layers, the hidden unit size, and regularization techniques. Adjustments to the learning rate or optimization algorithms could further improve training dynamics.

The confusion matrix in Figure 4 provides a more granular view of the model's classification performance. While the LSTM model correctly classifies the majority of cyberbullying categories, it struggles particularly with distinguishing between the "other" and "none" categories. This suggests that these two categories are similar in terms of feature distribution, which may lead to misclassifications. Further

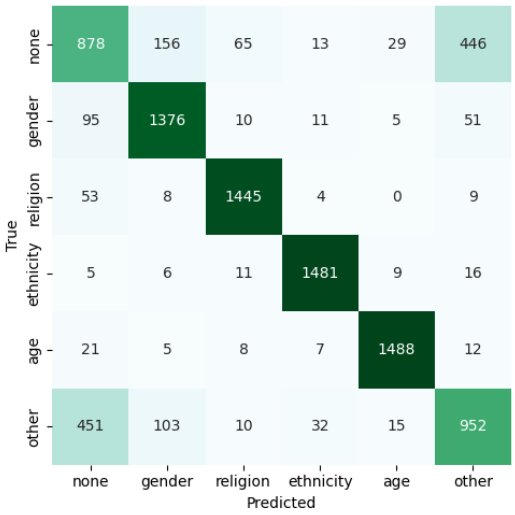

Figure 4: Confusion Matrix for RNN LSTM

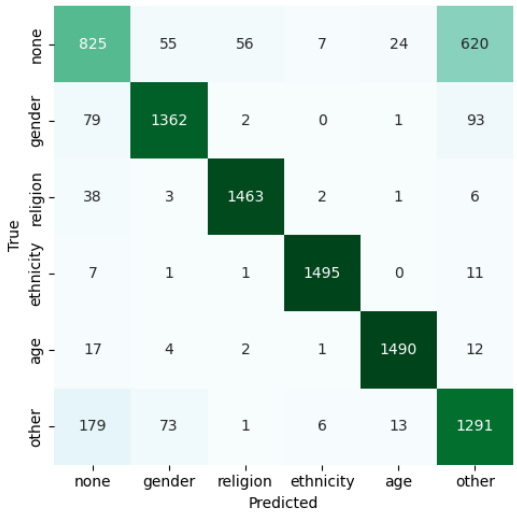

Figure 5: Confusion Matrix for BERT

refinements in the preprocessing pipeline, feature extraction, or model architecture could help the network to better differentiate between these subtle distinctions. Techniques such as data augmentation or using more sophisticated embedding methods for text representation could also improve model performance in these areas.

While the RNN LSTM model demonstrates solid overall performance, additional fine-tuning and refinements could further boost its ability to handle complex categorization challenges, especially between similar labels.

4.2. BERT Model Performance

Table 2
BERT Model Performance for Different LR and Epochs

LR	Epochs	Accuracy	Mean AUROC	Run Time
2e-5	2	85.78%	97.57%	27:01
2e-5	3	85.38%	97.39%	40:41
3e-5	2	84.91%	97.48%	27:16
3e-5	3	85.08%	97.43%	40:57

The best classification was selected based on these performance, prioritizing the accuracy of the model, but also choosing the model with lower Type I error than Type II error. I made this decision since there are many negative implications that comes with being wrongfully flagged for harmful opinions, such as demonetization of source of revenue, social image, and disregarding freedom of speech if it is not harmful. However, another reason was also since repeated offenses of cyberbullying will likely be detected even if there was to be a Type II error.

The confusion matrix shows a balanced classification, detecting cases where there is bullying correctly. The biggest error that is observed is within the 'other' categories, likely due to other forms of bullying may contain a large set of classes that make it difficult to distinguish them apart from none bullying context. This therefore predicts that nearly 40% of regular threads are being classified as bullying, leading to a large Type I error which is not ideal given the prior explanation.

Table 3
Classification Report for BERT

	precision	recall	f1-score	support
none	0.72	0.52	0.60	1587
gender	0.91	0.85	0.88	1537
religion	0.96	0.95	0.95	1513
ethnicity	0.98	0.99	0.98	1515
age	0.98	0.97	0.98	1526
other	0.62	0.84	0.71	1563
accuracy			0.85	9241
macro avg	0.86	0.85	0.85	9241
weighted avg	0.86	0.85	0.85	9241

When observing the classification report, this model has a good performance with most predictions, and there is room for improvement with the classification of other forms of bullying and non-bullying. It is notable that "gender", "religion", "ethnicity", and "age", are all classified very well with over 0.91 precision and over 0.85 recall. While the "other" category has a great recall score, the precision is very low, signifying that, like RNN-LSTM, it incorrectly labels a higher proportion of negative instances of "others" as positives.

4.3. RoBERTa Model Performance

RoBERTa was evaluated using multiple learning rates, epochs, and sequence lengths during testing. Ultimately, a maximum sequence length of 128 tokens (with padding) was chosen becasue of its superior performance and alignment with the dataset. As found in the literature review, RoBERTa generally outperforms BERT with larger training data, dynamic masking strategy, and increased batch size, despite dropping NSP task. However, given BERT's previously established high accuracy, the choice between

RoBERTa and BERT depends on the specific use case and context, which we will explore further in this subsection based on the results.

Table 4
BERT Model Performance for Different LR and Epochs

LR	Epochs	Accuracy	Mean AUROC	Run Time
2e-5	2	84.12%	97.45%	27:10
2e-5	3	85.25%	97.52%	40:45
3e-5	2	83.20%	97.18%	27:09
3e-5	3	84.83%	97.45%	40:44
5e-5	2	83.67%	97.07%	27:08
5e-5	3	84.43%	97.21%	40:43

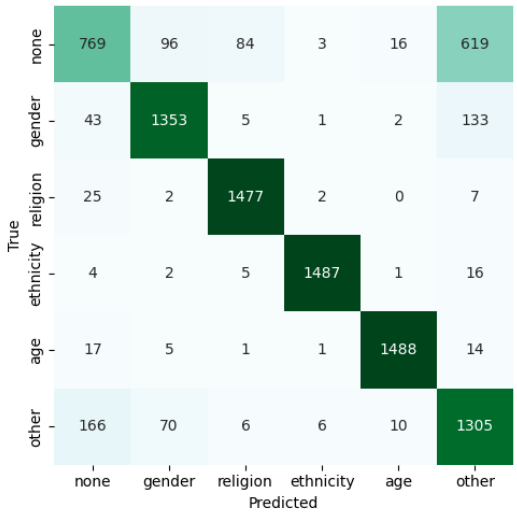

Figure 6: Confusion Matrix for RoBERTa

The RoBERTa results (Figure 6) face the same challenges as BERT but with better overall accuracy. There is slightly larger Type I error for the none-classification with identifying non-bullying threads as "others". Overall, it predicted less text as bullying in comparison.

Table 5
Classification Report for RoBERTa

	precision	recall	f1-score	support
none	0.75	0.48	0.59	1587
gender	0.89	0.88	0.88	1537
religion	0.94	0.98	0.96	1513
ethnicity	0.99	0.98	0.99	1515
age	0.98	0.98	0.98	1526
other	0.62	0.83	0.71	1563
accuracy			0.85	9241
macro avg	0.86	0.86	0.85	9241
weighted avg	0.86	0.85	0.85	9241

The classification table from Table 6 shows that when digging into each classification deeper, precision for "None" is better but recall is worse. The model is cautious and avoids falsely predicting "None," but it also fails to identify some true "None" cases. This trade-off contradicts our initial defined costs of false negatives and false positives. The recall and F1-score appears to be more balanced across the classes, while "None" and "others" still struggle to identify correctly.

5. Conclusion and Future Improvements

The handling of the "others" category in data preprocessing significantly impacted model performance. Testing both BERT and RoBERTa without the "others" classification resulted in an accuracy increase to over 95% for both models, particularly in classifying non-bullying tweets. This improvement is likely due to the class imbalance, which made it difficult for the models to learn from this category effectively. To address this, it would have been optimal to preprocess this class by either creating smaller subclasses or removing it entirely to mitigate the imbalance. Additionally, a common issue was the large running time, despite removing duplicates. The error here was dropping duplicates before text preprocessing, which led to some duplicates reappearing after preprocessing (likely due to emoji use, punctuation, and spacing errors). By removing duplicates during preprocessing, I was able to significantly reduce the running time with minimal impact on accuracy.

Both BERT and RoBERTa had higher accuracy than RNN, meaning that they accurately classify more frequently. Although RoBERTa was expected to outperform BERT, RoBERTa achieved a 97.52% accuracy with a learning rate of 2e-5 and 3 epochs, while BERT achieved a slightly higher accuracy of 97.57% under the same settings. Despite this, RoBERTa was more computationally expensive than BERT. In terms of classification, RoBERTa exhibited more balanced accuracy, precision, and recall across classes, whereas BERT showed a preference for minimizing Type I errors, which was desirable for our context. Interestingly, a simpler model such as the RNN with LSTM, regarding recall of "None", which is the type of error, outperformed both BERT and RoBERTa for that case. Overall, the fine-tuned transformer models had better and similar performances in this field, however, RNN with LSTM is able to compute these tasks relatively well with less computational power.

Other approaches that can be observed for fine tuning can be looking into other transformers-baed models or exploring Adapters and LoRA techniques that are designed to make these models more computationally efficient for fine-tuning. Adapters are lightweight modules added to the pre-trained models to adapt the model to a new task like cyberbully classification without changing the original model weights, inserting small set of parameters between the layers of pre-trained model [13]. LoRA applies low-rank decomposition to the weight matrices in transformer layers, enabling more efficient parameterization [14]. Both of these techniques provide a way to keep the core model frozen, reducing the computational load and improving the fine-tuning process on smaller, task-specific parameters, which can improve the classification of cyberbullying.

Moving forward, once a model is selected, the next step will be to implement it in production and continue

optimizing its performance to combat cyberbullying and form measures moving forward.

References

[1] Sudhanshu Kumar, Partha Pratim Roy, Debi Prosad Dogra, and Byung-Gyu Kim. A Comprehensive Review on Sentiment Analysis: Tasks, Approaches and Applications, November 2023. arXiv:2311.11250 [cs].

[2] Ralf C. Staudemeyer and Eric Rothstein Morris. Understanding LSTM – a tutorial into Long Short-Term Memory Recurrent Neural Networks, September 2019. arXiv:1909.09586 [cs].

[3] P.J. Werbos. Backpropagation through time: what it does and how to do it. *Proceedings of the IEEE*, 78(10):1550–1560, October 1990. Conference Name: Proceedings of the IEEE.

[4] Schmidhuber S. Hochreiter, S. Long Short-Term Memory. Neural computation, 9(8):1735–1780, 1997.

[5] S. Hochreiter. Untersuchungen zu dynamischen neuronalen Netzen.

[6] Ralf Staudemeyer and Eric Morris. [1909.09586] Understanding LSTM – a tutorial into Long Short-Term Memory Recurrent Neural Networks.

[7] Shuohang Wang and Jing Jiang. Learning Natural Language Inference with LSTM, November 2016. arXiv:1512.08849 [cs].

[8] Kuncahyo Setyo Nugroho, Anantha Yullian Sukmadewa, Haftittah Wuswilahaken DW, Fitra Abdurrachman Bachtiar, and Novanto Yudistira. BERT Fine-Tuning for Sentiment Analysis on Indonesian Mobile Apps Reviews. In *6th International Conference on Sustainable Information Engineering and Technology 2021*, pages 258–264, September 2021. arXiv:2107.06802 [cs].

[9] Tomas Mikolov, Ilya Sutskever, Kai Chen, Greg Corrado, and Jeffrey Dean. Distributed Representations of Words and Phrases and their Compositionality, October 2013. arXiv:1310.4546 [cs].

[10] Kusumaningrum R. Wibowo A. Muhammad, P. Sentiment Analysis Using Word2vec And Long Short-Term Memory (LSTM) For Indonesian Hotel Reviews. *ResearchGate*, October 2024.

[11] Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding, May 2019. arXiv:1810.04805 [cs].

[12] Yinhan Liu, Myle Ott, Naman Goyal, Jingfei Du, Mandar Joshi, Danqi Chen, Omer Levy, Mike Lewis, Luke Zettlemoyer, and Veselin Stoyanov. RoBERTa: A Robustly Optimized BERT Pretraining Approach, July 2019. arXiv:1907.11692 [cs].

[13] Neil Houlsby, Andrei Giurgiu, Stanislaw Jastrzebski, Bruna Morrone, Quentin de Laroussilhe, Andrea Gesmundo, Mona Attariyan, and Sylvain Gelly. Parameter-Efficient Transfer Learning for NLP, June 2019. arXiv:1902.00751 [cs].

[14] Edward J. Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. LoRA: Low-Rank Adaptation of Large Language Models, October 2021. arXiv:2106.09685 [cs].

[15] Z. Vujovic. Classification Model Evaluation Metrics. *ResearchGate*, October 2024.

[16] Syed Sihab-Us-Sakib, Md. Rashadur Rahman, Md. Shafiul Alam Forhad, and Md. Atiq Aziz. Cyberbullying detection of resource constrained language from social media using transformer-based approach. *Natural Language Processing Journal*, 9:100104, December 2024.

[17] T Fawcett. Introduction to receiver operating characteristics (roc) analysis.